

BISECTION MATLAB CODE

```
% Clearing Screen
clc

% Setting x as symbolic variable
syms x;

% Input Section
y = input('Enter non-linear equations: ');
a = input('Enter first guess: ');
b = input('Enter second guess: ');
e = input('Tolerable error: ');

% Finding Functional Value
fa = eval(subs(y,x,a));
fb = eval(subs(y,x,b));

% Implementing Bisection Method
if fa*fb > 0
    disp('Given initial values do not bracket the
root.');
```

```
else
    c = (a+b)/2;
    fc = eval(subs(y,x,c));
    fprintf('\n\na\t\tb\t\tc\t\tf(c)\n');
    while abs(fc)>e
        fprintf('%f\t%f\t%f\t%f\n',a,b,c,fc);
        if fa*fc< 0
            b =c;
        else
            a =c;
        end
        c = (a+b)/2;
        fc = eval(subs(y,x,c));
    end
    fprintf('\nRoot is: %f\n', c);
end
```

BISECTION PYTHON CODE

```
# Defining Function
def f(x):
    return x**3-5*x-9

# Implementing Bisection Method
def bisection(x0,x1,e):
    step = 1
    print('\n\n*** BISECTION METHOD IMPLEMENTATION ***')
    condition = True
    while condition:
        x2 = (x0 + x1)/2
        print('Iteration-%d, x2 = %0.6f and f(x2) = %0.6f' % (step, x2, f(x2)))

        if f(x0) * f(x2) < 0:
            x1 = x2
        else:
            x0 = x2

        step = step + 1
        condition = abs(f(x2)) > e

    print('\nRequired Root is : %0.8f' % x2)

# Input Section
x0 = input('First Guess: ')
```

```
x1 = input('Second Guess: ')
e = input('Tolerable Error: ')

# Converting input to float
x0 = float(x0)
x1 = float(x1)
e = float(e)

#Note: You can combine above two section like this
# x0 = float(input('First Guess: '))
# x1 = float(input('Second Guess: '))
# e = float(input('Tolerable Error: '))

# Checking Correctness of initial guess values and
bisecting
if f(x0) * f(x1) > 0.0:
    print('Given guess values do not bracket the
root.')
    print('Try Again with different guess values.')
else:
    bisection(x0,x1,e)
```